

1 bgc-argo-tutorials: A practical guide to biogeochemical 2 Argo data analysis

3 Haruto Fujishima ^{1*} and Hakase Hayashida ^{2*¶}

4 ¹ Department of Geophysics, Graduate School of Science, Tohoku University, Sendai, Japan ² Application
5 Laboratory, Japan Agency for Marine-Earth Science and Technology, Yokohama, Japan ¶ Corresponding
6 author * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Submitted: 16 April 2026

Published: unpublished

License

Authors of papers retain copyright¹³
and release the work under a ¹⁴
Creative Commons Attribution 4.0⁵
International License ([CC BY 4.0](#)).

7 Summary

8 Biogeochemical Argo (BGC-Argo) is an ocean observing system for monitoring the health of
9 marine ecosystems by a global array of autonomous profiling floats that routinely measure
10 biogeochemical oceanographic properties in the upper 2,000 m of the water column every
11 10 days or so ([Claustre et al., 2020](#)). By 2030, the global BGC-Argo community aims to
12 cover the global ocean with 1,000 floats equipped with sensors that can measure up to
13 six key variables: chlorophyll-a, pH, oxygen, nitrate, irradiance, and suspended particles
14 ([Thierry et al., 2025](#)). The collected profiles are made publicly available in the netCDF
15 format within a day or so, and can be used for scientific analysis and operational purposes.

16 Here we introduce **bgc-argo-tutorials**, which provides a step-by-step guide on the analysis
17 of BGC-Argo data using Python and Jupyter Notebook. This learning module is designed
18 for graduate students and early-career researchers of Oceanography (e.g., Early Career
19 Ocean Professionals or ECOPs; [Li et al., 2025](#)) ready to dive into BGC-Argo data analysis
20 for their research projects. A basic knowledge on the Argo program ([Wong et al., 2020](#)) and
21 BGC-Argo ([Bittig et al., 2019](#)) is a prerequisite. **bgc-argo-tutorials** provides lessons on
22 how to search, identify, download, and post-process the profiles for conducting time-series
23 analysis. The end product of **bgc-argo-tutorials** (gridded 2D time-depth arrays) can
24 be used for understanding the spatial and temporal variability of ocean dynamics and
25 biogeochemistry, evaluating the performance of numerical model simulations and other
26 data products, and improving ocean biogeochemical prediction. **bgc-argo-tutorials** aims
27 to help expand the end users of the growing BGC-Argo data in Japan and beyond.

28 Statement of need

29 While the BGC-Argo database is an invaluable resource, the native formatting of the
30 NetCDF files presents a steep learning curve for new users. Direct utilization of the raw
31 datasets is computationally cumbersome due to non-standard data typing, such as the
32 use of string arrays for Quality Control (QC) flags instead of easily maskable integers.
33 Furthermore, because the vertical sampling depths vary irregularly across individual
34 profiles, the data must be interpolated onto a standardized depth grid to enable continuous
35 2D visualizations and direct comparisons with gridded model and other data products.
36 **bgc-argo-tutorials** addresses these bottlenecks by providing working examples and
37 functions for the necessary post-processing of the data.

38 The scope and approach of **bgc-argo-tutorials** are structurally different from other
39 existing softwares, learning modules, and data products related to BGC-Argo, because it
40 requires users to understand the source code for each function developed within (cf. [Maze
41 & Balem, 2020](#)), it focuses specifically on BGC-Argo and is targeted at users with basic

knowledge on BGC-Argo (cf. [González-Santana & Vélez-Belchí, 2024](#)), and it gives access to post-processed data on a near-real-time basis (cf. [Johnson et al., 2023](#)). The idea for **bgc-argo-tutorials** was conceived when the corresponding author started using the BGC-Argo data for the first time and experienced the issues mentioned above ([Hayashida et al., 2025](#)). The author also noticed the relatively low use cases by the Japanese researchers (out of the 849 BGC-Argo papers published, only 24 papers are led by authors with the Japanese surnames; <https://argo.ucsd.edu/outreach/publications/biogeochemical-argo-bibliography/> as of February 2, 2026). To address this regional issue, we provide support for **bgc-argo-tutorials** in Japanese and organize on-site tutorial sessions in the near future.

Earlier versions of **bgc-argo-tutorials** were presented at the Japan Geoscience Union Meeting 2025 in Chiba and the Oceanographic Society of Japan Fall Meeting 2025 in Hokkaido. A hands-on workshop on BGC-Argo data analysis using **bgc-argo-tutorials** will be held as a side event of the OneArgo international workshop in Miyagi on June 2, 2026. An earlier version of **bgc-argo-tutorials** has been used for the assessment of seasonal ocean prediction of chlorophyll-a and nitrate in the equatorial Pacific by a global climate model ([Doi et al., submitted](#)).

Acknowledgements

BGC-Argo data are made freely available through the two Global Data Assembly Centres (GDACs; [SEANOE, 2026](#)). We thank Hidehiro Fujio for computational assistance and Pete Strutton, Clara Vives, Kanako Sato, Yoshimi Kawai for insightful discussion on BGC-Argo floats and feedback on earlier versions of **bgc-argo-tutorials**. This work was supported by JSPS KAKENHI Grant Number JP24H02226 and the Nakajima Foundation.

References

- Bittig, H. C., Maurer, T. L., Plant, J. N., Schmechtig, C., Wong, A. P. S., Claustre, H., Trull, T. W., Udaya Bhaskar, T. V. S., Boss, E., Dall’Olmo, G., Organelli, E., Poteau, A., Johnson, K. S., Hanstein, C., Leymarie, E., Le Reste, S., Riser, S. C., Rupan, A. R., Taillandier, V., ... Xing, X. (2019). A BGC-Argo Guide: Planning, Deployment, Data Handling and Usage. *Frontiers in Marine Science*, 6. <https://doi.org/10.3389/fmars.2019.0050>
- Claustre, H., Johnson, K. S., & Takeshita, Y. (2020). Observing the Global Ocean with Biogeochemical-Argo. *Annual Review of Marine Science*, 12, 23–48. <https://doi.org/10.1146/annurev-marine-010419-010956>
- Doi, T., Baba, Y., & Hayashida, H. (submitted). Seasonal chlorophyll-a prediction in the tropical pacific with a global climate model incorporating marine biogeochemistry (SINTEX-F2bio). *Journal of Climate*.
- González-Santana, A., & Vélez-Belchí, P. (2024). The Argo Online School: An e-learning tool to get started with Argo. *Journal of Open Source Education*, 7(80), 193. <https://doi.org/10.21105/jose.00193>
- Hayashida, H., Strutton, P. G., Hashihama, F., Horimoto-Miyazaki, N., Katano, T., & Yasui-Tamura, S. (2025). Status and prospects of biogeochemical Argo observations of chlorophyll-a in the northwest Pacific. *Journal of Oceanography*, 81, 289–299. <https://doi.org/10.1007/s10872-025-00750-2>
- Johnson, K. S., Fassbender, A., Gray, A., Nicholson, D., Purkey, S., Riser, S. C., Takeshita, Y., Talley, L. D., Wijffels, S. E., Gilson, J., Grady, L. A., Guisewhite, N., Maurer, T. L., Parise, K., Plant, J. N., Robbins, P., Rupan, R. A., & Swift, D. D. (2023). *Southern*

- 88 *Ocean Carbon and Climate Observations and Modeling (SOCCOM) and Global Ocean*
89 *Biogeochemistry (GO-BGC) Biogeochemical-Argo Float Data Archive*. UC San Diego
90 Library Digital Collections. <https://doi.org/10.6075/JOSJ1KT8>
- 91 Li, S., Addey, C. I., Roman, R., Hayashida, H., Jiang, C., Hu, C., Coronado-Álvarez, L.
92 de L. A., Lim, H.-G., Akmal, S. G., Orji, C. M., Arora, P., Li, R., Pm, S., Adesina, R.
93 B., Lindemann, C., Ma, D., Sarkar, S., Mascioni, M., Monteiro, T., ... Luo, T. (2025).
94 Early career ocean professionals' declaration on ocean negative carbon emissions for our
95 ocean and future. *The Innovation*, 101007. <https://doi.org/10.1016/j.xinn.2025.101007>
- 96 Maze, G., & Balem, K. (2020). Argopy: A Python library for Argo ocean data analysis.
97 *Journal of Open Source Software*, 5(53), 2425. <https://doi.org/10.21105/joss.02425>
- 98 SEANOE (Ed.). (2026). *Argo float data and metadata from global data assembly centre*
99 *(argo GDAC)*. <https://doi.org/10.17882/42182>
- 100 Thierry, V., Claustre, H., Pasqueron de Fommervault, O., Zilberman, N., Johnson, K.
101 S., King, B. A., Wijffels, S. E., Bhaskar, U. T. V. S., Balmaseda, M. A., Belbeoch,
102 M., Bollard, M., Boutin, J., Boyd, P., Cancouët, R., Chai, F., Ciavatta, S., Crane,
103 R., Cravatte, S., Dall'Olmo, G., ... Zuo, H. (2025). Advancing ocean monitoring and
104 knowledge for societal benefit: The urgency to expand Argo to OneArgo by 2030.
105 *Frontiers in Marine Science*, 12. <https://doi.org/10.3389/fmars.2025.1593904>
- 106 Wong, A. P. S., Wijffels, S. E., Riser, S. C., Pouliquen, S., Hosoda, S., Roemmich, D.,
107 Gilson, J., Johnson, G. C., Martini, K., Murphy, D. J., Scanderbeg, M., Bhaskar, T. V.
108 S. U., Buck, J. J. H., Merceur, F., Carval, T., Maze, G., Cabanes, C., André, X., Poffa,
109 N., ... Park, H.-M. (2020). Argo Data 1999–2019: Two Million Temperature-Salinity
110 Profiles and Subsurface Velocity Observations From a Global Array of Profiling Floats.
111 *Frontiers in Marine Science*, 7, 700. <https://doi.org/10.3389/fmars.2020.00700>